

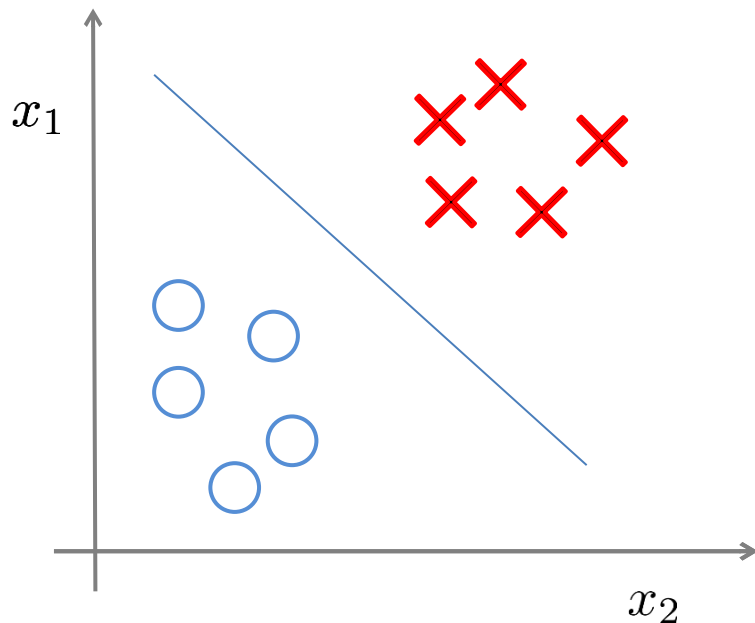
K-Means & DBSCAN Algorithm

Artificial Neural Networks

R Bharath

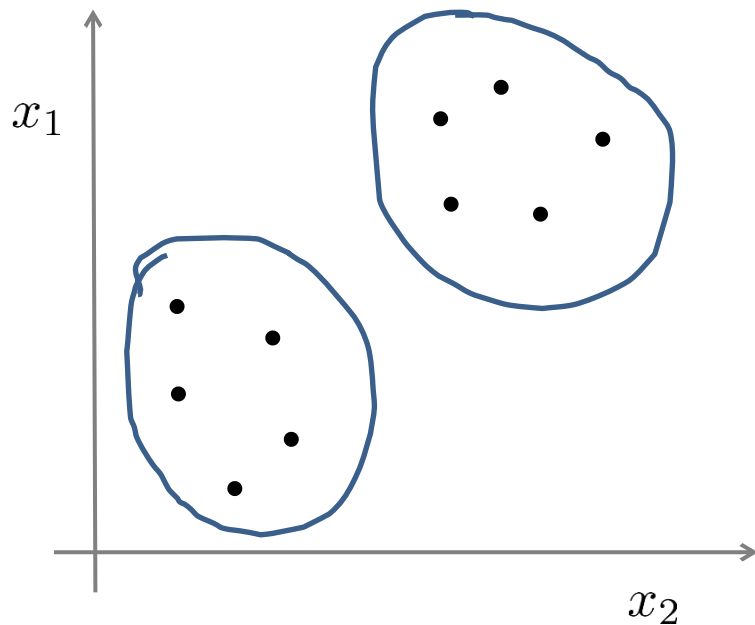


Supervised learning



Training set: $\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), (x^{(3)}, y^{(3)}), \dots, (x^{(m)}, y^{(m)})\}$

Unsupervised learning

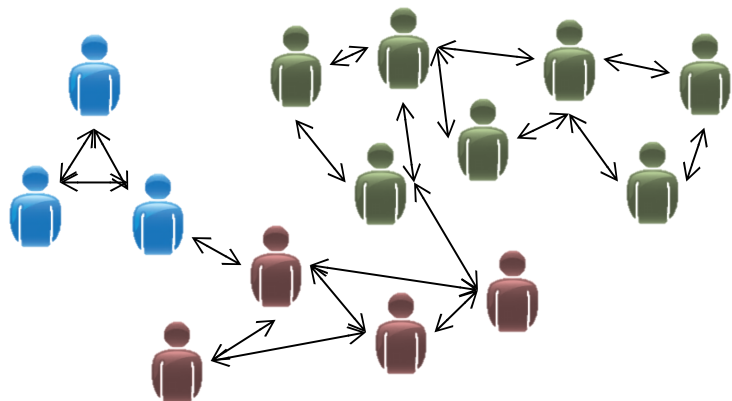


Training set: $\{x^{(1)}, x^{(2)}, x^{(3)}, \dots, x^{(m)}\}$

Applications of clustering



Market segmentation



Social network analysis

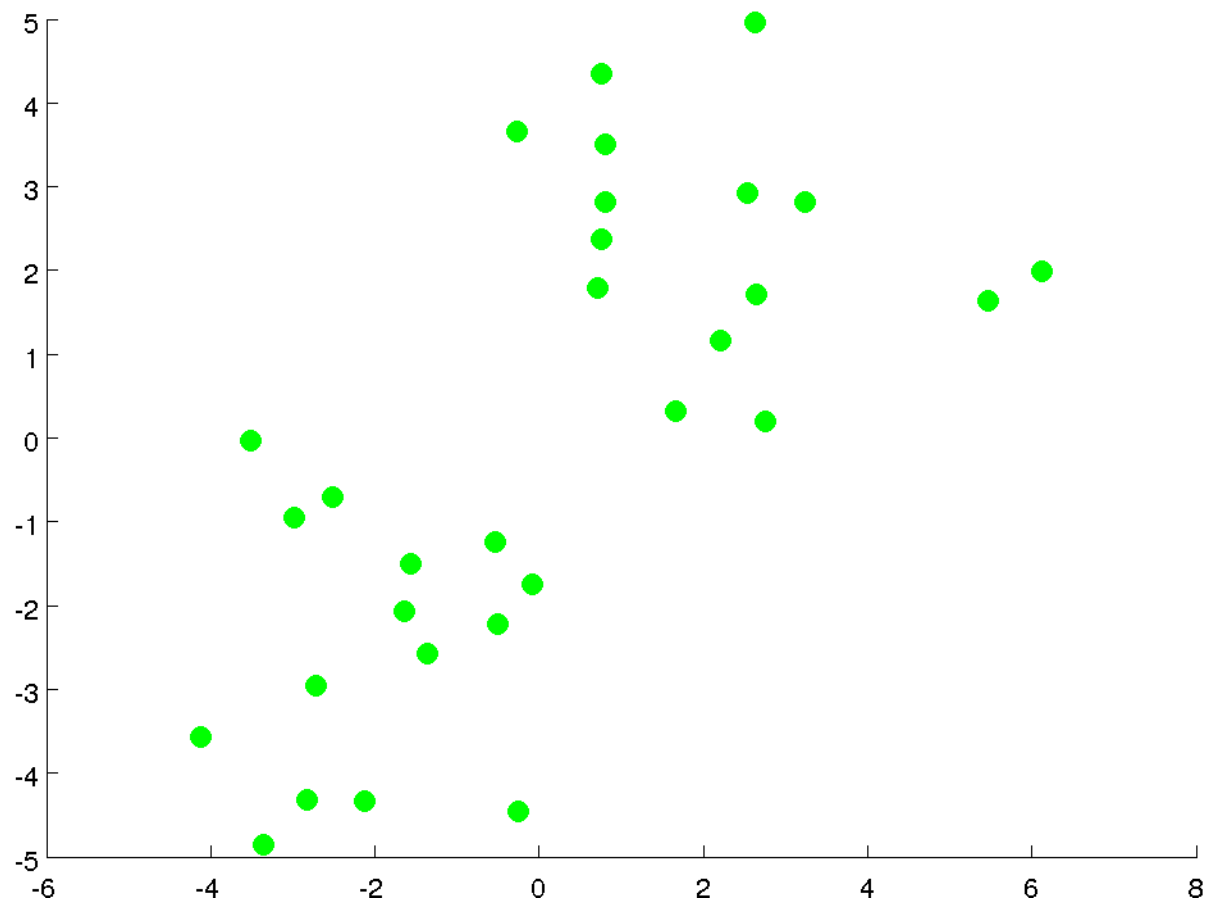


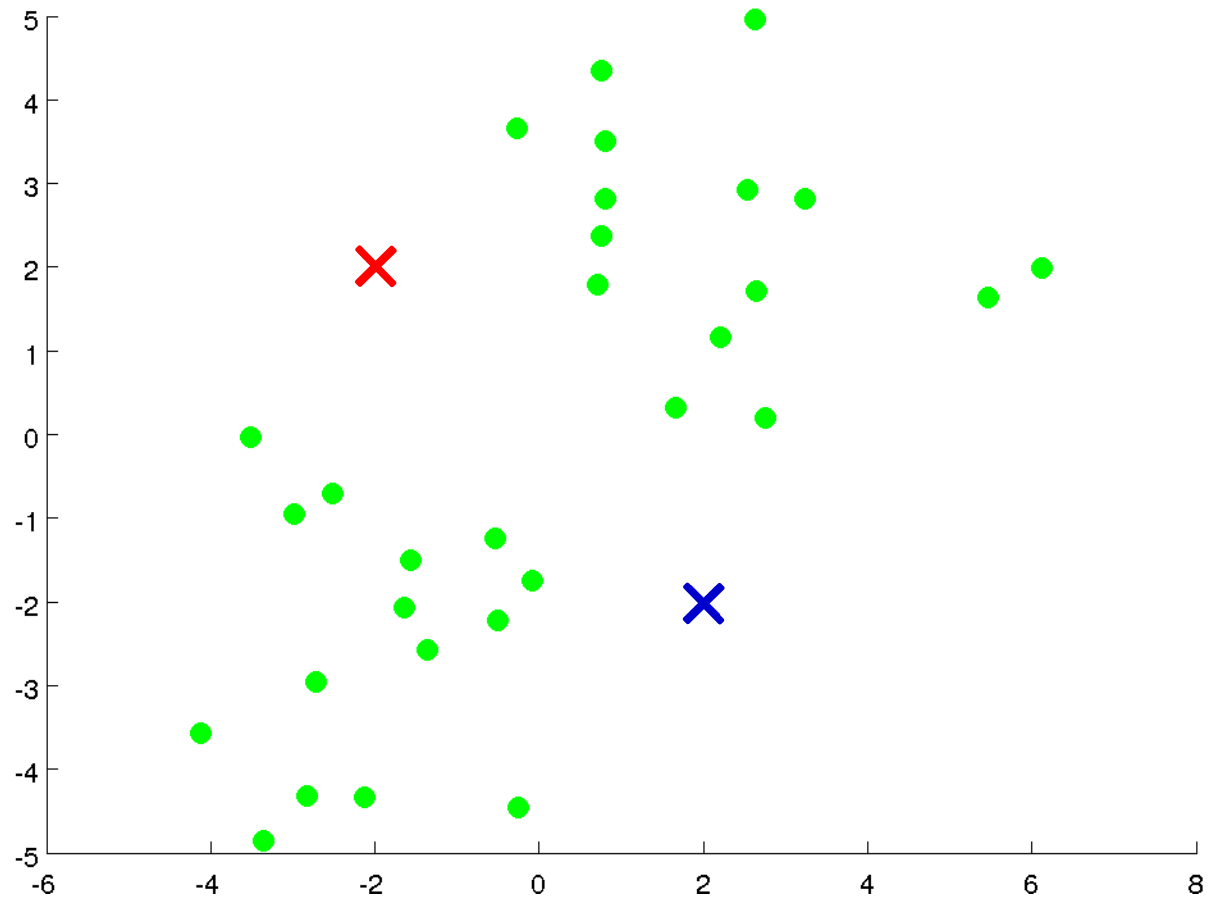
Organize computing clusters

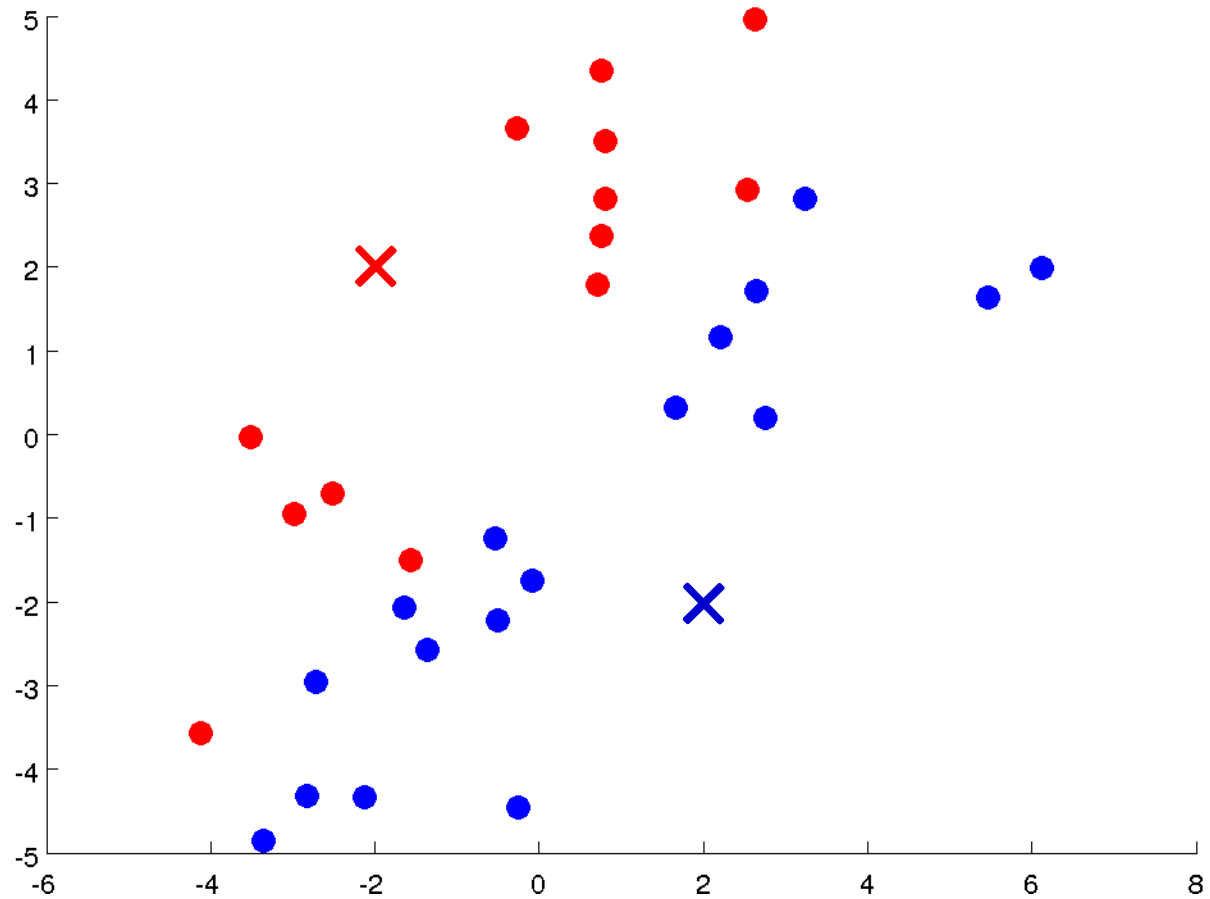


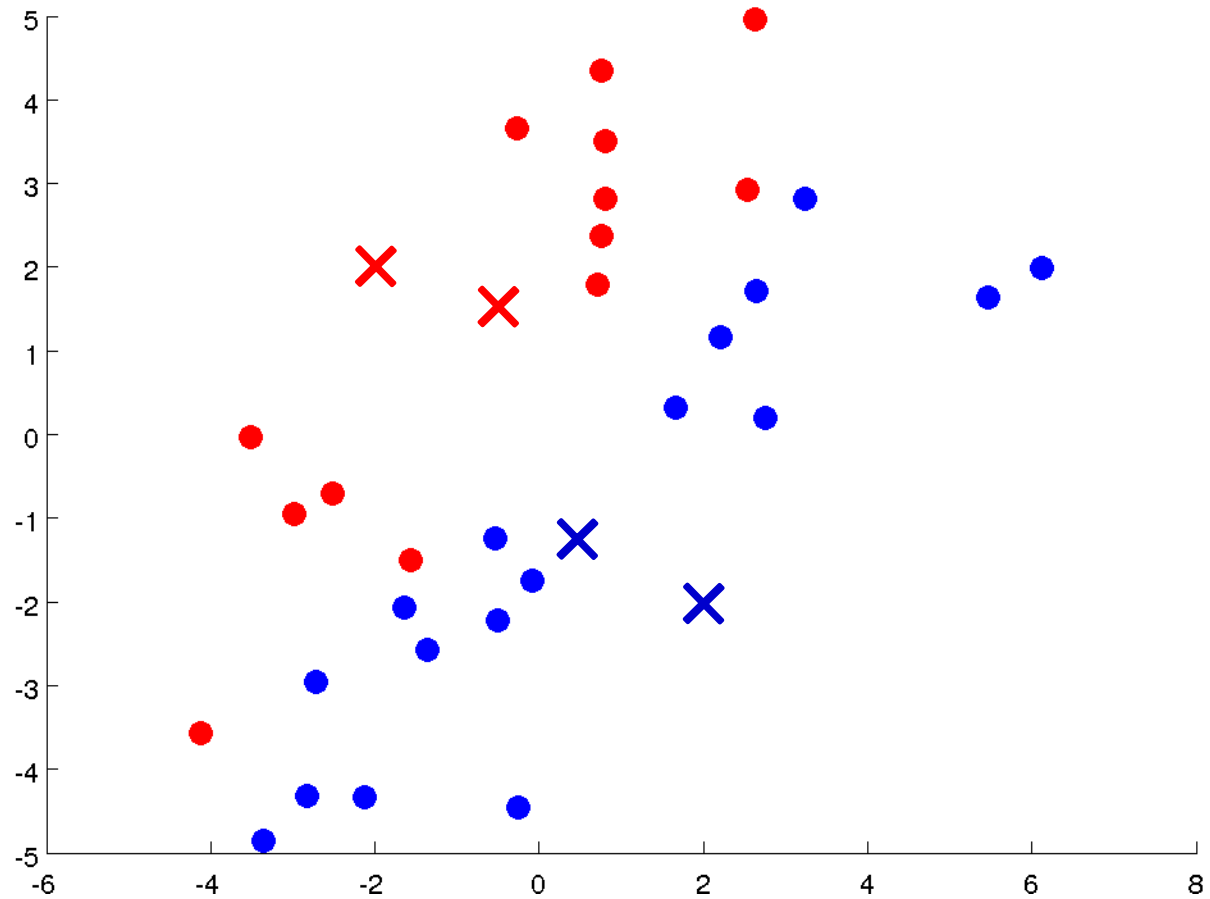
Image credit: NASA/JPL-Caltech/E. Churchwell (Univ. of Wisconsin, Madison)

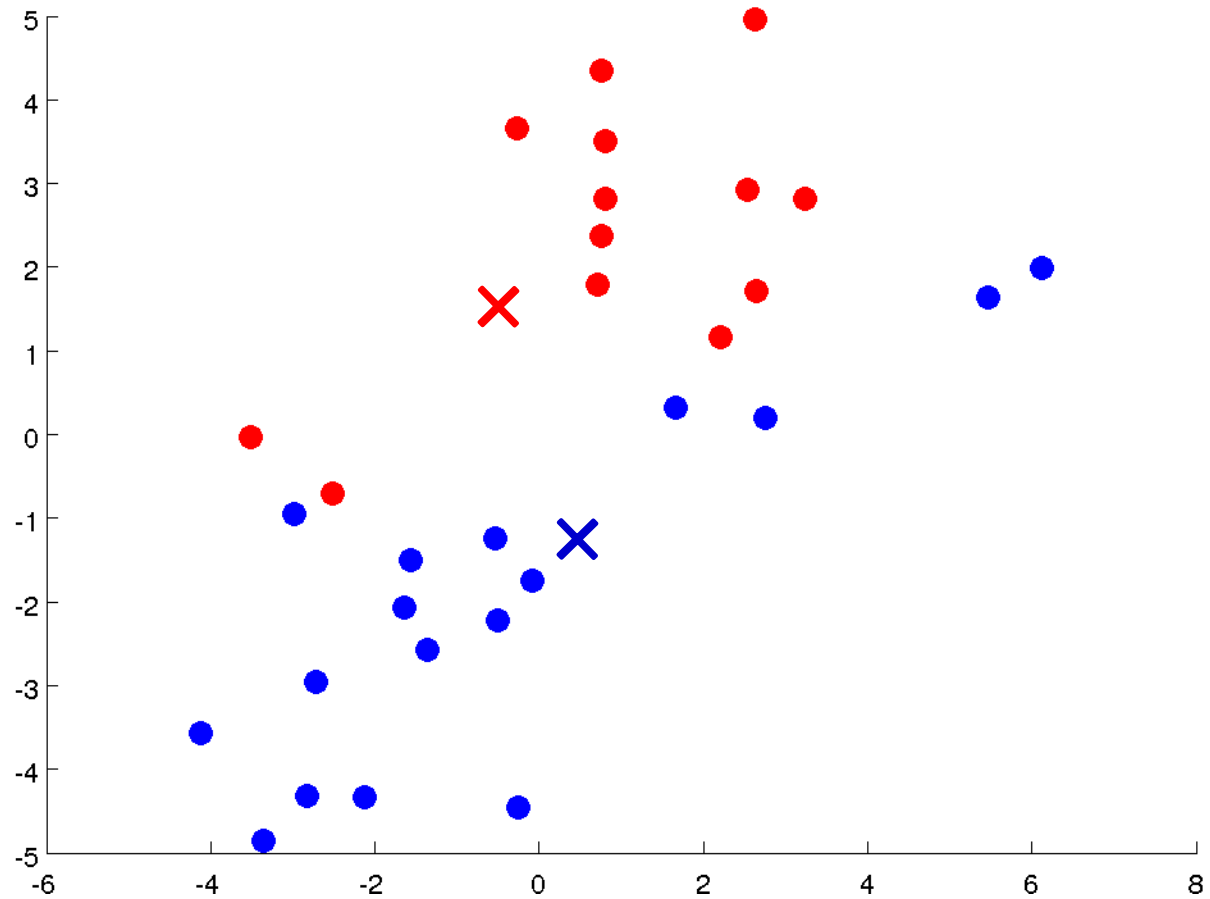
Astronomical data analysis

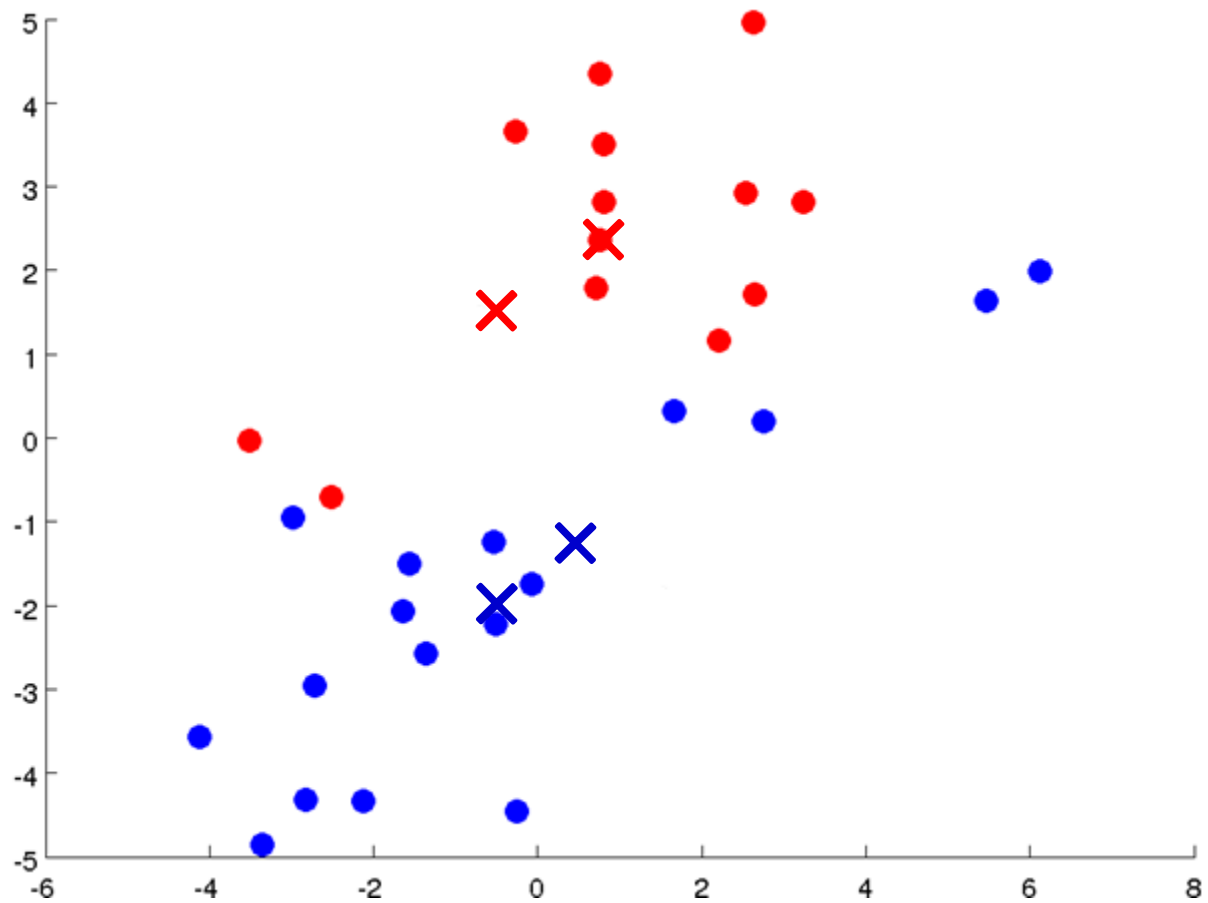


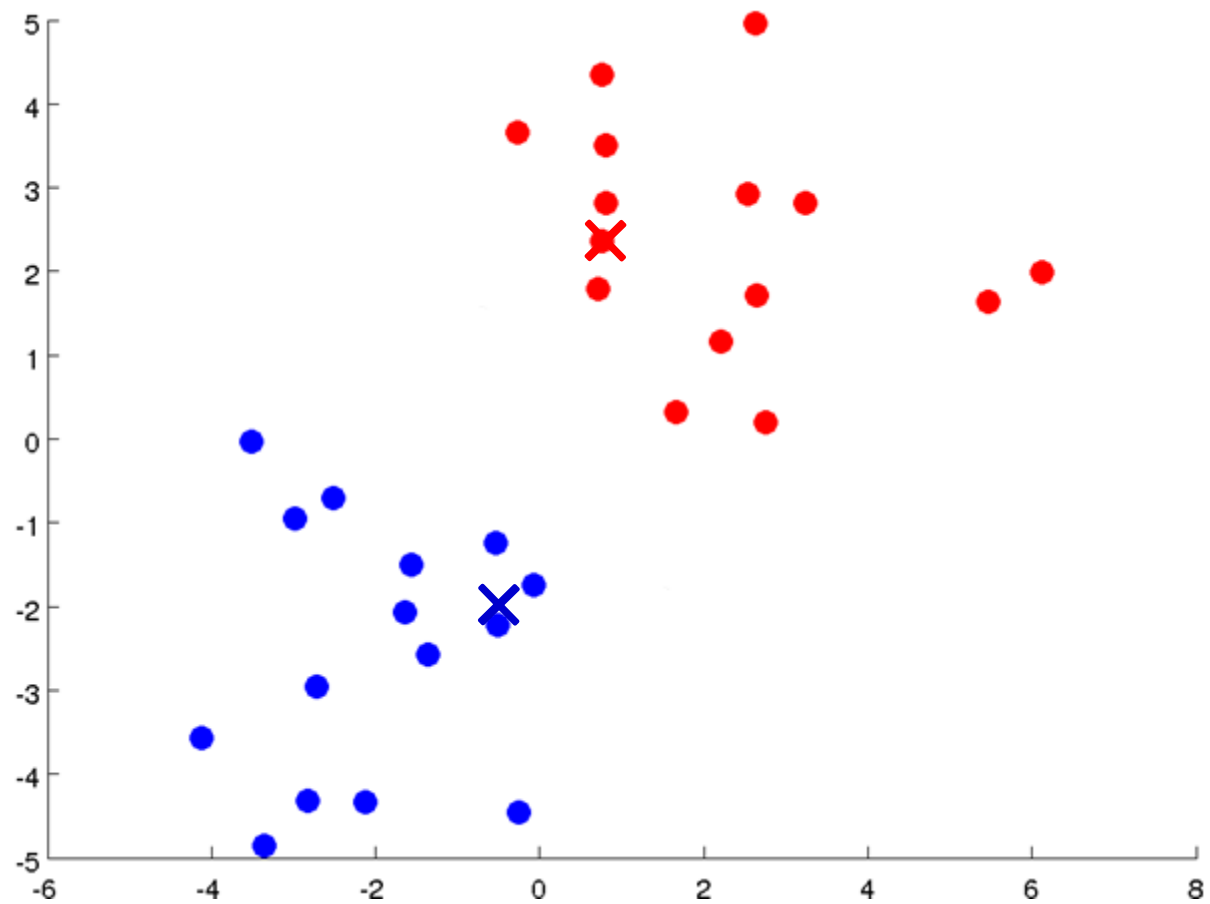


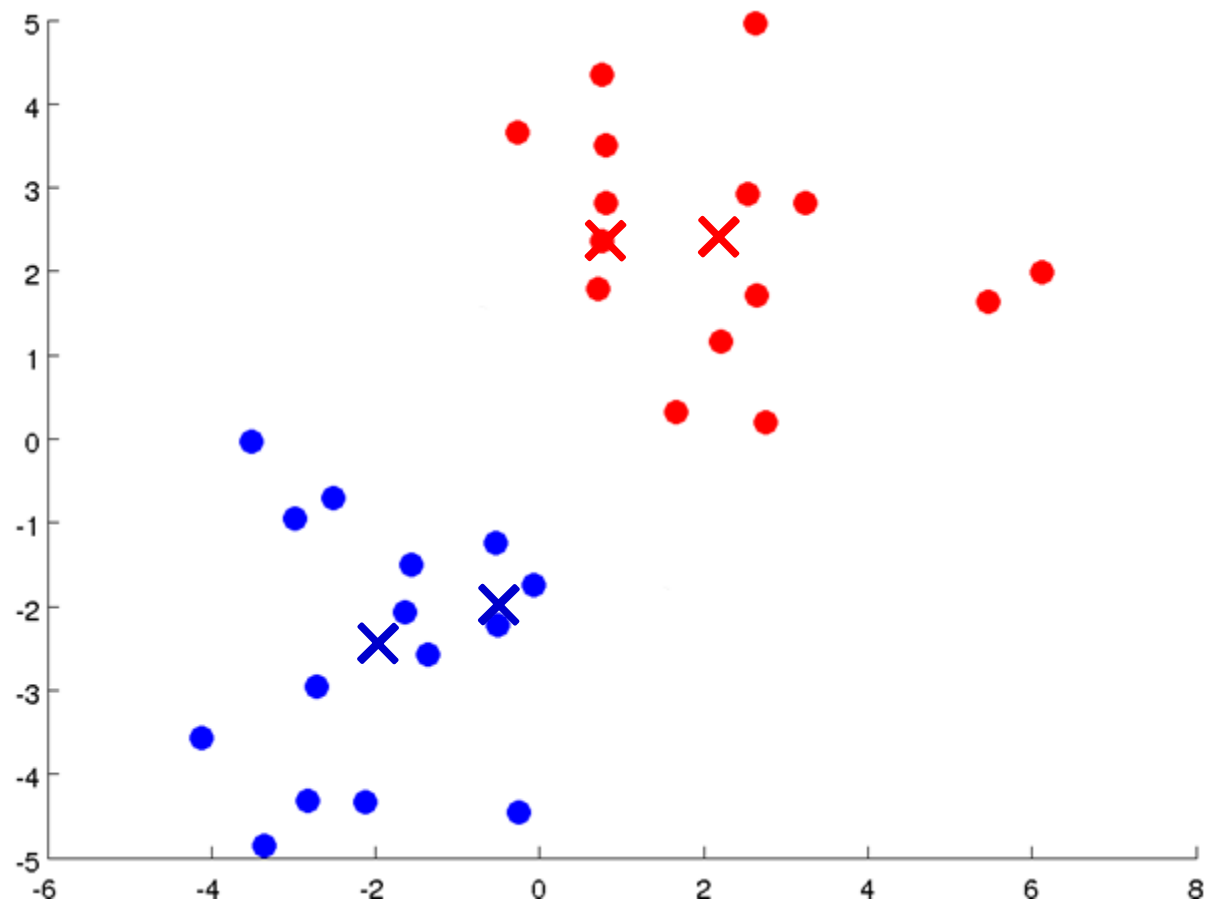


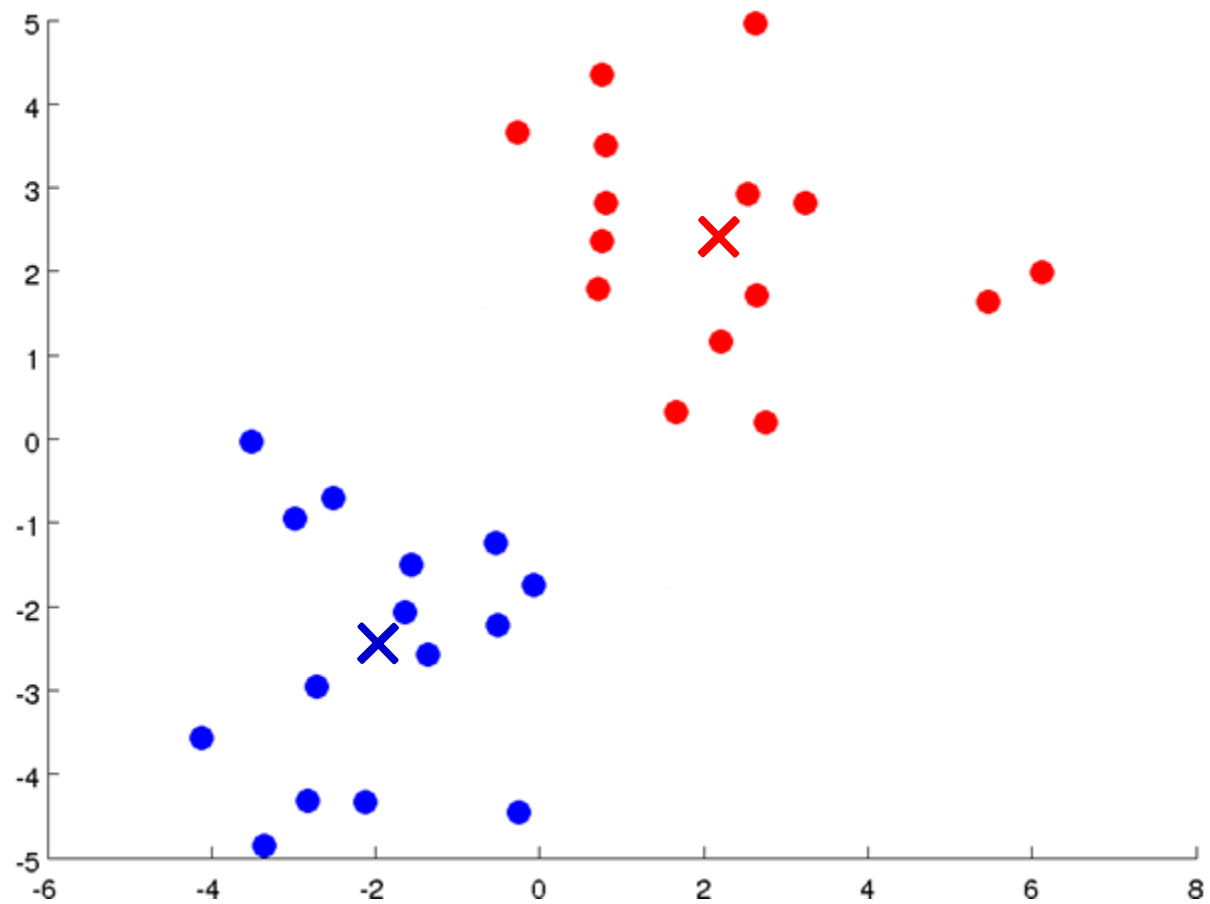












K-means algorithm

Randomly initialize K cluster centroids $\mu_1, \mu_2, \dots, \mu_K \in \mathbb{R}^n$

Repeat {

 for $i = 1$ to m

$c^{(i)} :=$ index (from 1 to K) of cluster centroid
 closest to $x^{(i)}$

 for $k = 1$ to K

$\mu_k :=$ average (mean) of points assigned to cluster k

}

Drawbacks

- Need to specify the Cluster Size
- Sensitive to Noise
- The performance varies with the initialization

k-means



DBSCAN

Density based Spatial Clustering of Applications
in presence of Noise

Working of DBSCAN

DBSCAN algorithm requires two parameters:

- 1. eps:** It defines the neighborhood around a data point i.e. if the distance between two points is lower or equal to 'eps' then they are considered neighbors.
- 2. MinPts:** Minimum number of neighbors (data points) within eps radius.

As a general rule, the minimum MinPts can be derived from the number of dimensions D in the dataset as, $\text{MinPts} \geq D+1$.

Density-based spatial clustering of applications with noise (DBSCAN)

- **Algorithmic steps for DBSCAN clustering**
 - The algorithm proceeds by arbitrarily picking up a point in the dataset (until all points have been visited).
 - If there are at least 'minPoint' points within a radius of ' ϵ ' to the point then we consider all these points to be part of the same cluster.
 - The clusters are then expanded by recursively repeating the neighborhood calculation for each neighboring point

About Data Points

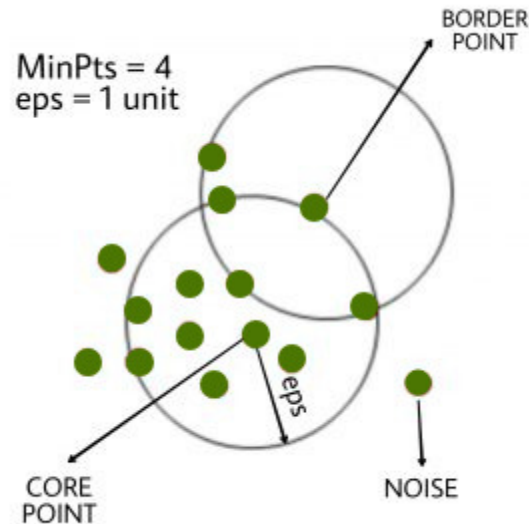
In this algorithm, we have 3 types of data points.

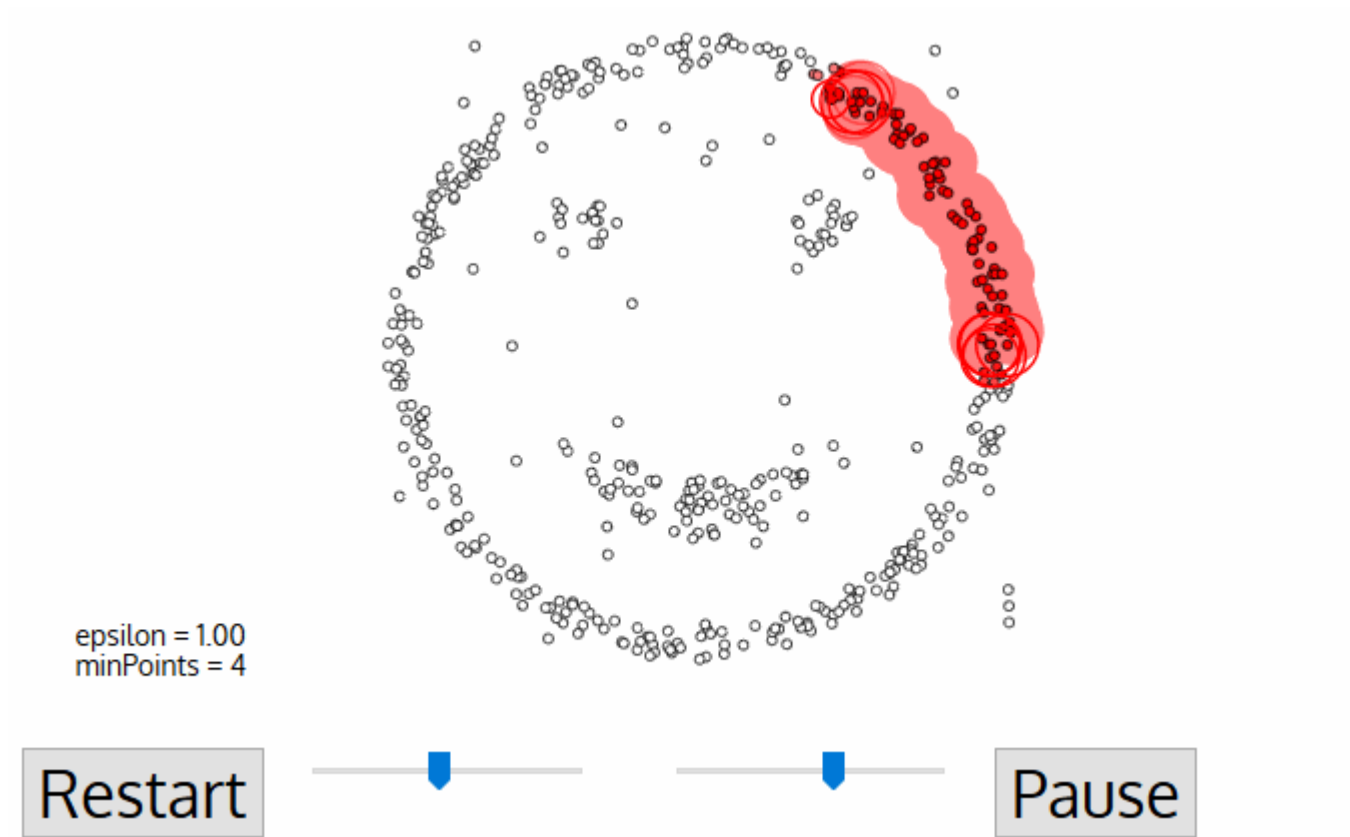
Core Point: A point is a core point if it has more than $MinPts$ points within eps .

Border Point: A point that has fewer than $MinPts$ within eps but it is in the neighborhood of a core point.

Noise or outlier:

A point that is not a core point or border point.





DBSCAN()

DBSCAN



k-means

